Spacing San Francisco:
Examining Inequity in Access to Essential Amenities

Group 9: Dynamic Duo!
Brittany Vang
Taylor Vassar

Track A

May 12th, 2026

1. Introduction

Essential resources are unevenly distributed in urban environments, often reflecting underlying socioeconomic inequalities. This project asks: *How are key neighborhood amenities distributed across socioeconomically different areas of San Francisco, and to what extent do these spatial patterns contribute to the compounded disadvantage?*

We define amenities as essential services that support daily life and long-term opportunity, and specifically examine food access, public transit, and public schools. These resources are foundational to meeting an individual's biological and sociological needs: to feed oneself, travel, and gain an education. The project draws on data from 2020 to 2025 and uses the census tract as its unit of analysis, enabling neighborhood-level comparisons while remaining compatible with socioeconomic data sources.

Crucially, we move beyond surface-level mapping and simple point counts, recognizing that not all amenities carry equal weight. To capture this complexity, we apply a weighting schema grounded in accessibility literature, distinguishing, for example, between food vendors by type and transit stops by route frequency and destination quality. This motivates three sub-questions:

1. **Food:** Which tracts have lower-quality food access when food options are weighted by type and service level?
2. **Transit:** Which tracts have worse effective transit access when stops are weighted by route frequency and destination quality?
3. **Schools:** Which tracts are more likely to score poorly across a combined food, transit, and school accessibility index simultaneously?

The metrics used include accessibility indicators (distance-based and network-based), amenity density, and composite indices measuring "amenity richness" or deprivation. Key stakeholders include low-income residents, urban planners, policymakers, and organizations focused on urban equity. Spatial disparities in resource access can entrench poverty traps and reduce social mobility, a concern with significant analytical and practical consequences.

2. Literature Review

Existing research establishes that deprivation is not an individual phenomenon but a cumulative one. In The Case of Montréal's Missing Food Deserts[1], the authors argue that a status of deprivation cannot be attributed to any single characteristic, but more often to an accumulation of many, a framing that directly motivates our multi-amenity index approach. This case also shows the limits of relying on simple amenity counts: food deserts that seem absent in overall data may become visible when access is measured using weighted or network-based methods. This concern aligns with broader critiques of incomplete or misleading datasets discussed in Dark Data, which emphasizes how what is omitted from measurement can distort conclusions about inequality and access[2].

[1] Apparicio P, Cloutier MS, Shearmur R. The case of Montréal's missing food deserts: evaluation of accessibility to food supermarkets. Int J Health Geogr. 2007 Feb 12;6:4.
[2] HAND, DAVID J. Dark Data: Why What You Don't Know Matters. Princeton University Press, 2020. https://doi.org/10.2307/j.ctvmd85db.

Research focused on San Francisco further shows that access to food is not evenly distributed across the city. Even when grocery stores may appear nearby on a map, lower-income neighborhoods can still face real barriers to accessing healthy and affordable food[3]. This points to the need for accessibility measures that reflect real-world travel conditions rather than idealized proximity. These disparities are especially important given the role of food assistance programs such as Supplemental Nutrition Assistance Program (SNAP) in mitigating food insecurity across underserved communities[4].

The transit literature similarly motivates our weighting approach. Research on transit-dependent populations shows that unequal access to transit can deepen existing inequalities[5]. People with access to a car often have far more opportunities than those who rely only on public transportation, which means the quality and connectivity of transit service matters just as much as whether transit exists at all[6]. Stops with infrequent service or limited destinations provide functionally different access than well-connected nodes, and treating them as equivalent would obscure meaningful disparities.

Finally, research on school location and funding shows that educational opportunity is shaped by more than just how close a student lives to a school. Differences in school quality and funding often reflect broader neighborhood inequalities, making access to strong educational resources an important part of the compounded disadvantage this project seeks to measure[7]. Together, this body of work supports our central methodological choice: understanding urban inequality requires moving from simple amenity counts toward weighted, multi-dimensional accessibility indices that reflect how resources are actually experienced by residents.

3. Data

Census: The census tract dataset outlines subdivisions of San Francisco County based on how census data is collected, as of 2020. The income dataset contains the 2020 - 2024 five-year estimates of these individual census tracts. The census raw data had a few issues: the geometry was stored as plain text instead of a usable map format, the ID fields used to join the two datasets didn't match because of extra characters in the ACS/income dataset, and income values were stored as text rather than numbers. There was also an empty column with no values. To fix these, the geometry was converted and reprojected to a coordinate system suited for San Francisco (ESPG: 2227), the IDs were standardized so the datasets could be joined correctly, columns were cleaned up and renamed, and a couple of offshore island tracts (the Farallons) were removed since they skewed the map. No duplicate records were found, the joined dataset came out complete with no missing values, and the final cleaned file was saved as a GeoJSON for use in the rest of the project.

Food: The Active Food Services dataset documents various types of eateries (restaurants, cafés, etc.) in different neighborhoods in San Francisco. Some of these services we will not consider in our analysis due to its restriction to openness to the public (i.e. employee-only cafeterias and bars). The SNAP Retailers dataset indicates places that accept SNAP as a form of payment. Lastly, the Mobile Food Services dataset locates food trucks and push carts. The common issue with the raw data was

[3] Ramirez, Yesenia, "Food Insecurity in San Francisco (Examination of some local and state interventions)" (2022). Master's Projects and Capstones. 1485.
[4] USDA. 2025. “Supplemental Nutrition Assistance Program (SNAP) | USDA-FNS.” Usda.gov.
[5] Maharjan, S., Janatabadi, F., & Ermagun, A. (2024). Spatial Inequity of Transit and Automobile Access Gap across America for Underserved Population. Transportation Research Record: Journal of the Transportation Research Board, 2678(1), 674-690.
[6] Akbar, Prottoy A. 2024. "Public Transit Access and Income Segregation." CEPR Discussion Paper 19558.
[7] Zheng, Angela, and James Graham. 2022. "Public Education Inequality and Intergenerational Mobility." American Economic Journal: Macroeconomics 14 (3): 250–82.

that it included locations that had expired permits or no longer existed. Exact duplicates were dropped, and a majority of the data cleaning involved removing the locations that were not active in the relevant time frame.

Transit: The transit dataset, which focuses on Muni, details the system's stopping locations and frequency. After attempting to cross check any inconsistencies with the given data and eliminate exact duplicate stops that were *not* for detour use, we found that the extent of our cleaning was limited to excluding metadata columns that were not relevant for our analysis.

Schools: Public school location data for the 2024–25 academic year was sourced from the Institute of Education Sciences, which provides the name, address, and coordinates of all 132 public schools in San Francisco. The raw schools data was a national dataset, so the first step was filtering it down to San Francisco County only, which left 132 schools. Columns were then renamed for clarity and unnecessary fields were dropped. The cleaned dataframe was converted to a GeoDataFrame using the latitude and longitude columns, reprojected to match the SF tracts coordinate system (EPSG:2227), and confirmed to have no missing values before being saved as a GeoJSON. Finally, we calculated schools per census tract and distances from tract centroids to the nearest school.

Table 1. Data Sources and Description

Data & Description	Coverage	Granularity	Type	Format	Source
Census Tracts	San Francisco County, 2020	Citywide-Level, Multipolygon Coordinates, Individual Tracts	Continuous	CSV	DataSF
Income	San Francisco County, 2020 - 2024	Median Value of Individual Census Tracts	Discrete	CSV	US Census Bureau
Active Food Services	San Francisco, 1960 - Present	Citywide-Level, Individual Businesses	Categorical	CSV	DataSF
SNAP Retailers	San Francisco, 2020	Citywide-Level, Individual Businesses, Addresses	Categorical	CSV	PolicyMap
Mobile Food Services	San Francisco, 2015 - Present	Citywide-Level, Individual Businesses, Addresses	Categorical	CSV	Open Data SF
Muni Stops	San Francisco County, 2025	Citywide-Level, Latitude & Longitude, Individual Stops	Mixed[8]	CSV	DataSF
School Locations	San Francisco County, 2024-5	Citywide-Level, Latitude & Longitude	Continuous	CSV	NCES

Note: *All original and modified versions of the data are available on our Github.*

Missing Data

Below, we summarize the missing information from each dataset (M), its connection to *Dark Data* (D), and how we are managing its absence (A). After considering *all* of the Dark Data types, we feel that those highlighted below describe the circumstances of the missing data the best.

[8] Contains categorical, discrete, and continuous data.

Census + Income

- **M:** After merging the census boundary layer with the median income dataset, we found that 93% of median income information for census tracts was missing.
- **D:** When merging both, tracts with small populations were likely removed to protect privacy and treated as NaN (Type 1: Data We Know Are Missing). In addition, certain populations such as undocumented or unhoused residents are likely undercounted or missing (Type 4: Self Selection/Non-Response).
- **A:** At this time, we left them as NaNs and colored grey so as to not make assumptions about suppressed census data.

Food

- **M:** The Active Food services dataset contained missing values about the location's end date, address information, and business corridor. The SNAP Retailers dataset interestingly contained no missing values. Lastly, the Mobile Food Services dataset had some locations missing their facility type, expiration date, days and hours of operation, and the type of food sold.
- **D:** Across the board, we removed columns that were irrelevant for our analysis (Type 1: Data We Know Is Missing), filtered for specific types of locations that aligned with our objective (Type 3: Data Cases That Are Specifically Chosen), and acknowledged that the accessibility and qualities of these locations have evolved over time (Type 7: Data That Changes With Time). A prominent example are locations that initially did not accept EBT as payment, but now do.
- **A:** These actions described above drastically reduced the amount of missing values in the datasets. The remaining absences are not influential, but were thoroughly examined; for example, a missing value under the business date expiration column is satisfactory because we are only interested in locations that are still open anyways. However, if the latitude and longitude for a location was missing, we used the available address to geocode and find those values.

Transit

- **M:** There is missing data for a few metadata columns.
- **D:** The columns that were excluded from analysis were also those with missing values, so they had little impact (Type 1: Data We Know Is Missing). Muni networks have expanded greatly since its inception in 1912, which *implies* greater access to various neighborhoods (Type 2: Data We Don't Know Is Missing & Type 7: Data That Changes With Time).
- **A**: As an example, for the missing values in the accessibilitymask variable, we kept them as-is, since the documentation did not elaborate on what it meant and we cannot determine the real-life traits of those stops. If the entire dataset was used, we would cross-check with another reputable source about the Muni Stops and impute the information from there.

Schools

- **M:** None.
- **D:** While our school location dataset was cleaned without any missing data, the data only consists of public schools, leaving out private and charter schools (Type 3: Data Specifically Chosen). In addition, the data solely contains location information, limiting how much can be inferred about other measures of access (Type 5 Missing Important Unobserved Aspects).
- **A:** N/A. If there were missing data, they would have been dropped.

4. Methods & Approach

This project looks at how access to essential neighborhood resources varies across San Francisco's 197 census tracts and whether lower-income neighborhoods experience overlapping disadvantages in food, transit, and school access. We use census tracts as neighborhood-scale units because they align well with demographic data from the American Community Survey, making it possible to compare amenity access with measures like median household income[9]. Using data from DataSF, PolicyMap, and NCES, we map food services and SNAP retailers, Muni transit stops, and public schools across the city. In our analysis, we use both simple counts of amenities and weighted accessibility measures to better capture how residents actually experience access in daily life. For example, food locations are differentiated by type and service quality, while transit stops are weighted by route frequency and connectivity. School access is measured using both counts and distance to the nearest school. These measures are then combined into a broader accessibility index to identify neighborhoods facing multiple forms of disadvantage at once. Finally, we use correlation analysis and spatial mapping to explore how these accessibility patterns relate to neighborhood income and broader socioeconomic inequality across the city.

5. Results and Key Findings

Median Income by Census Tracts: **Figure 1** reveals stark income inequality across San Francisco's census tracts, ranging from $18,217 in the poorest tract to $247,500 in the wealthiest. Wealthier areas cluster in the north while lower-income tracts concentrate in the center and southeast. This income divide is the foundation for the analysis: if amenities like food access, transit, and schools follow the same spatial pattern, lower-income neighborhoods may face compounded disadvantage rather than isolated gaps.

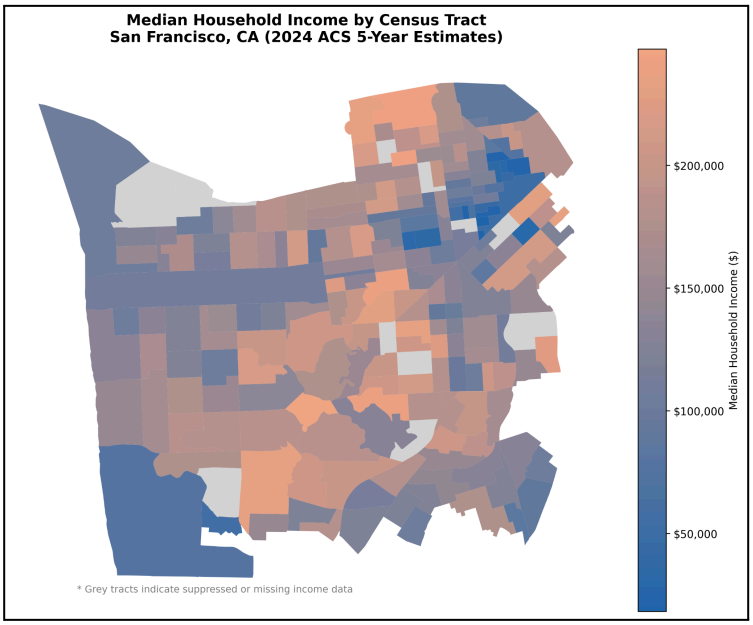

Figure 1. Distribution for estimated median income in US Dollars for San Francisco households, aggregated by census tract.

[9] Krieger, Nancy. 2006. "A Century of Census Tracts: Health & the Body Politic (1906–2006)." Journal of Urban Health 83 (3): 355–61.

<u>Food:</u> Each dataset showed an interesting pattern. For the Active Food Services (**Figure 2**), retail markets, caterers that also allow for dine-in service, and take-out facilities are the top three categories. This makes sense due to the fact that the remaining categories progressively become more limited in traits such as food inventory or scheduling.

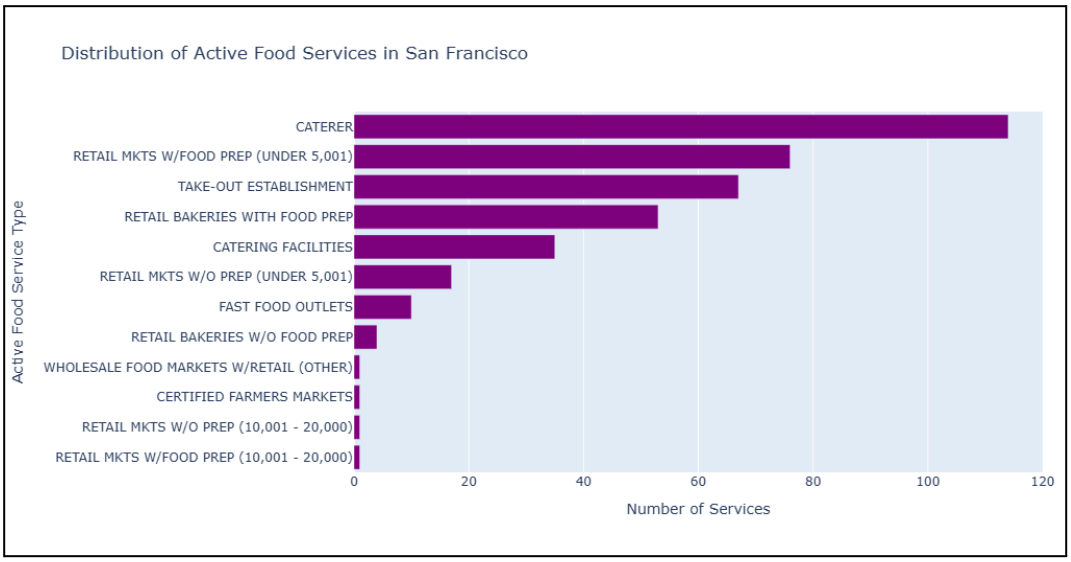

Figure 2. Distribution of Active Food Services by category.

In terms of geographic distribution, **Figure 3** below shows that a majority of the eateries lie in the northeast cluster of San Francisco. Furthermore, these locations are mostly within census tracts that observe less than $50,000 of medium income. Moving south, there is another cluster of eateries where the median income lies in the $100,000 - $120,000 range. However, we see the opposite phenomenon on the west side, where median income typically exceeds $100,000 but varies widely from then on.

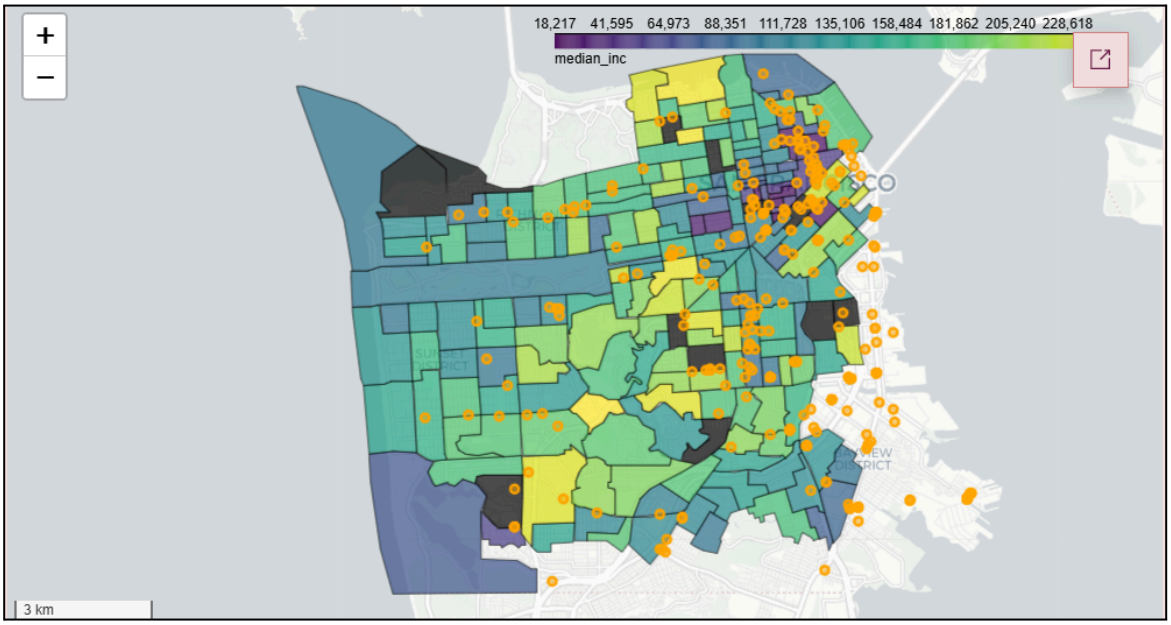

Figure 3. Map of Active Food Services in San Francisco.
https://taylorvassar.github.io/Spacing_San_Francisco/visualizations/interactive/food_services_income_map.html

Eateries on this part of the county are far more spread out; as you move away from Golden Gate Park (Census Tract 9803/the rectangle in the northwest corner) in the north and south direction they become prominent again. This seems typical since a park would tend to have more take-out and smaller eatery type establishments since people are more on-the-go. For the food portion of the observational analysis, these types of eateries will hold the **most weight** due to their commonality and eclectic nature.

Figure 4 below shows that for SNAP retailers, convenience stores construct the majority of retailers that accept SNAP as payment. However, convenience stores usually do not provide *full* meals; it is mostly to-go items or snacks. Additionally, the remaining types of retailers primarily serve as a vendor to collect ingredients for meals, rather than a place to get a ready-made meal. Lastly, SNAP regulations restrict a majority of recipients to only be able to purchase a certain type of meal (i.e. a non-senior SNAP recipient would not be able to purchase a warm slice of pizza from Whole Foods with SNAP, but a cold sandwich would be allowed). This is especially true for the Restaurant Meals Program.

Because of the high variability in the type of SNAP retailer and bureaucratic rules of the program, SNAP retailers will have **medium weight** in our analysis.

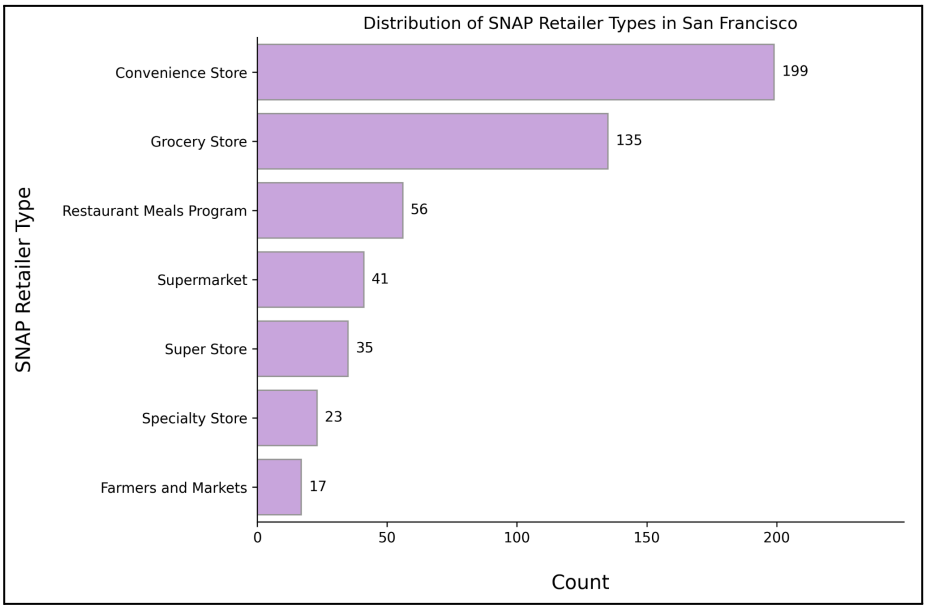

Figure 4. Distribution of SNAP Retailers in San Francisco.

Lastly, **Figure 5** shows the map of mobile food services, which are food trucks and push carts with an active permit. Although these are popular sources among on-the-go pedestrians, this type recorded the lowest numbers in our observational analysis. Because of this, we will assign it a **low weight**.

A majority of these sites are clustered in the northeast area of the county. Interestingly, they are dispersed over census tracts that represent the full range of median incomes. Several others span outside the census tracts that we have income information for, so this unknown data may skew our analysis.

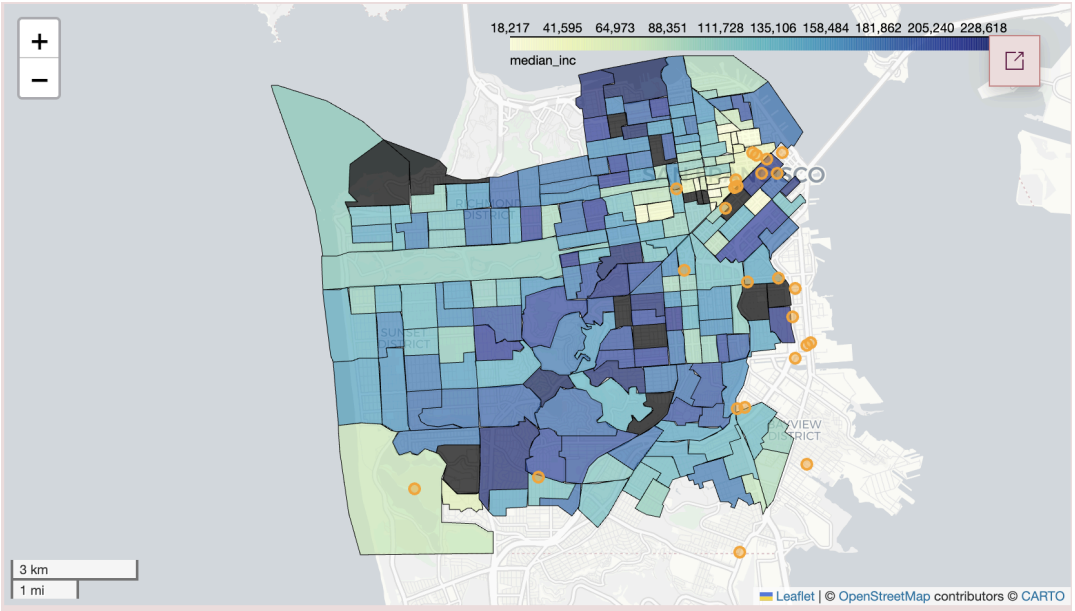

Figure 5. The distribution of Mobile Food Services (food trucks and push carts) displayed over the census tracts of varying socioeconomic status.
https://taylorvassar.github.io/Spacing_San_Francisco/visualizations/interactive/mobile_food_map.html

Overall, based on our sample, food services with a high weight are densely populated and span across all socioeconomic tracts on the east side of the county, whereas the west side observes a more sparse pattern among medium-income tracts. Food services with a medium weight are led by convenience and grocery stores, implying that the resource of SNAP itself may be the determinant of one's income level, rather than the census tract that their residence lies in. However, due to difficulty in generating an interactive map that displays these SNAP services, we cannot draw firm conclusions. Lastly, for the resources with a low weight, they seem to be concentrated in tracts that are at either extreme end of the income spectrum. This may reflect the income background of the mobile food servicers themself, or desire to gain income by selling in high-income areas.

Transit: The distribution of Muni transit stops (**Figure 6**) reveals dense coverage across the northeastern quadrant of San Francisco, particularly in downtown and the Financial District, where stops cluster heavily along a tight grid pattern. Coverage thins noticeably toward the southwestern areas, suggesting lower transit frequency in those neighborhoods. A small cluster of outlier stops in the upper right likely represents service extending into Marin County or the northern waterfront, sitting spatially separate from the main network.

When transit stops are overlaid on the median income layer (**Figure 7**), the relationship is not straightforward. Middle income central tracts benefit from the densest stop coverage, while the lowest income southeastern tracts like Bayview Hunters Point show both lower income and fewer stops, pointing to a compounded disadvantage. This pattern partially supports our research question, suggesting that transit access does not simply follow income citywide, but lower income neighborhoods in the southeast may face the most acute gaps in coverage.

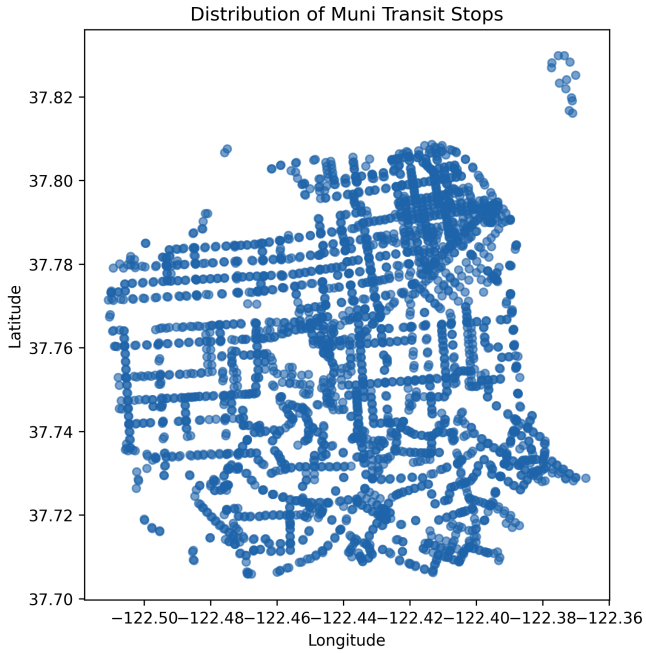

Figure 6. Distribution of Muni transit stops in SF.

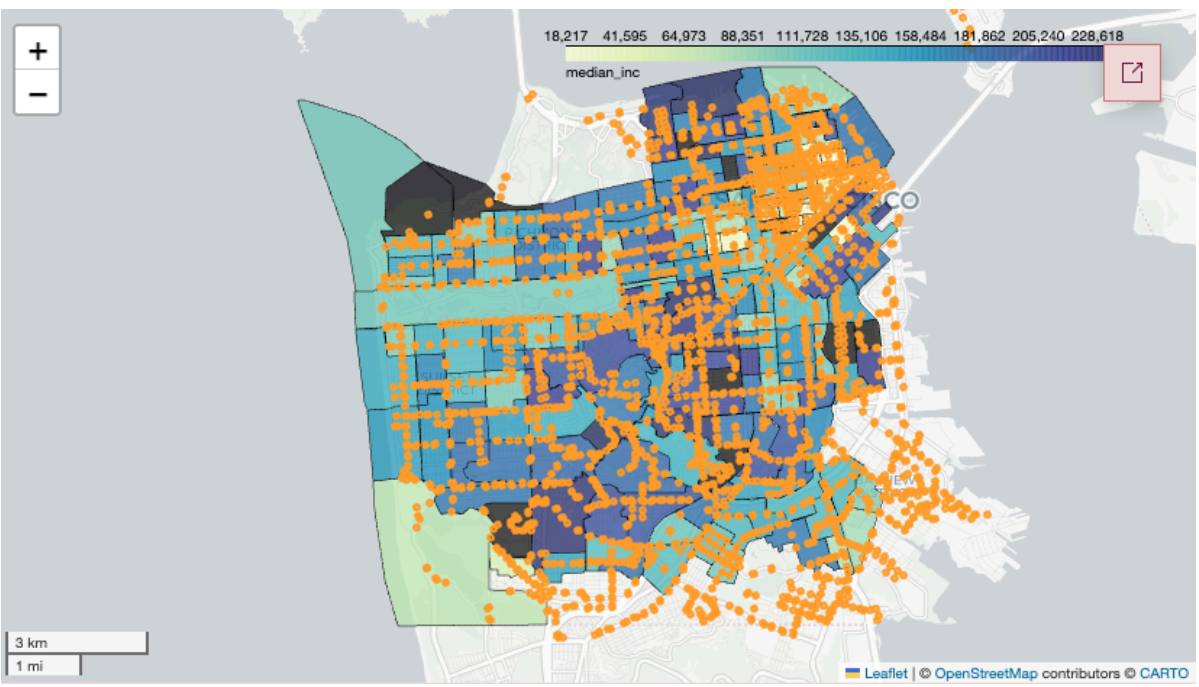

Figure 7. The distribution of Muni stations displayed over the census tracts of varying socioeconomic status. https://taylorvassar.github.io/Spacing_San_Francisco/visualizations/interactive/transit_map.html

Schools: On average, census tracts are about 405 meters from the nearest public school, and half of all tracts are within 366 meters which is roughly a 5-minute walk, suggesting that public school

access is generally pretty good across most of San Francisco **(Figure 8)**. However, the map tells a more nuanced story in which the northwest corner has tracts over 2,000 meters from the nearest school, shown in dark red, while the school point map confirms that schools are much more densely clustered in the central and eastern parts of the city. The wide range from 28 meters to 2,146 meters suggests that while most residents are well served, a small number of tracts particularly in the west and southwest are significantly farther from public schools than the rest of the city. **Figure 9** confirms that darker northwest tracts have no nearby school points, while central and eastern areas are densely served. Interestingly, some southern tracts near Bayview show moderate school proximity despite lower income levels, suggesting school access may not be the primary driver of disadvantage there. This uneven pattern across amenity types is exactly what our composite index aims to capture.

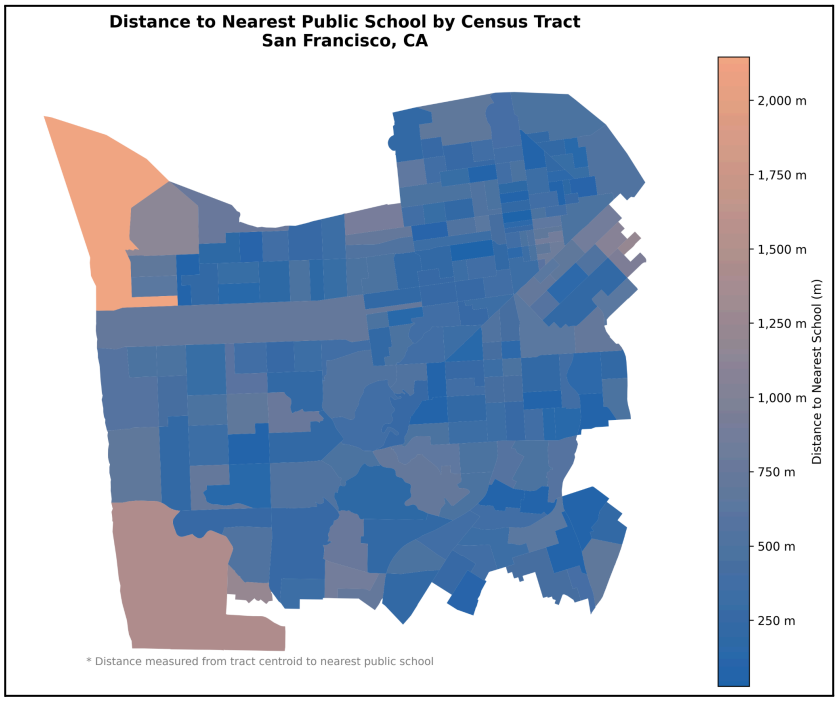

Figure 8. Distance to nearest public school by census tract, San Francisco, CA.

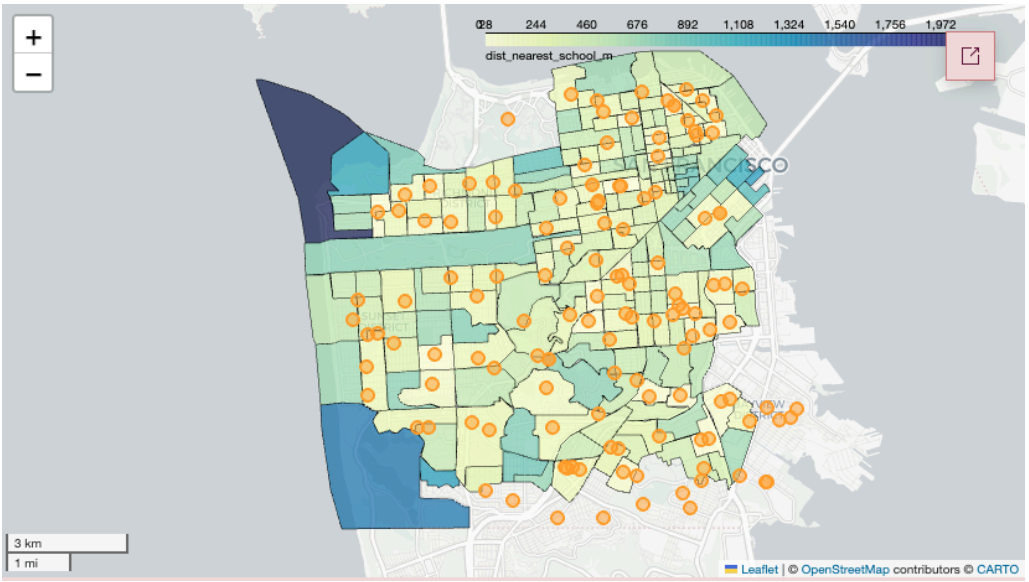

Figure 9. Distribution of public schools across San Francisco's census tracts (2024–25). https://taylorvassar.github.io/Spacing_San_Francisco/visualizations/interactive/school_map.html

Correlation and Composite Findings: The correlation analysis reveals that transit access has a weak positive relationship with income (r = 0.20), meaning higher income tracts tend to have slightly more stops, while food services and school proximity show virtually no linear relationship with income (r = -0.06 each). The composite index map reinforces this, showing that the worst-served tracts in orange cluster in the northwest and parts of the southwest, regions that also tend toward lower or suppressed income data. Together these findings offer a partial answer to our research question: while income alone does not strongly predict access to any single amenity, the spatial concentration of low composite scores in specific neighborhoods suggests that some tracts do experience overlapping gaps across food, transit, and schools simultaneously, which is the compounded disadvantage our index was designed to detect.

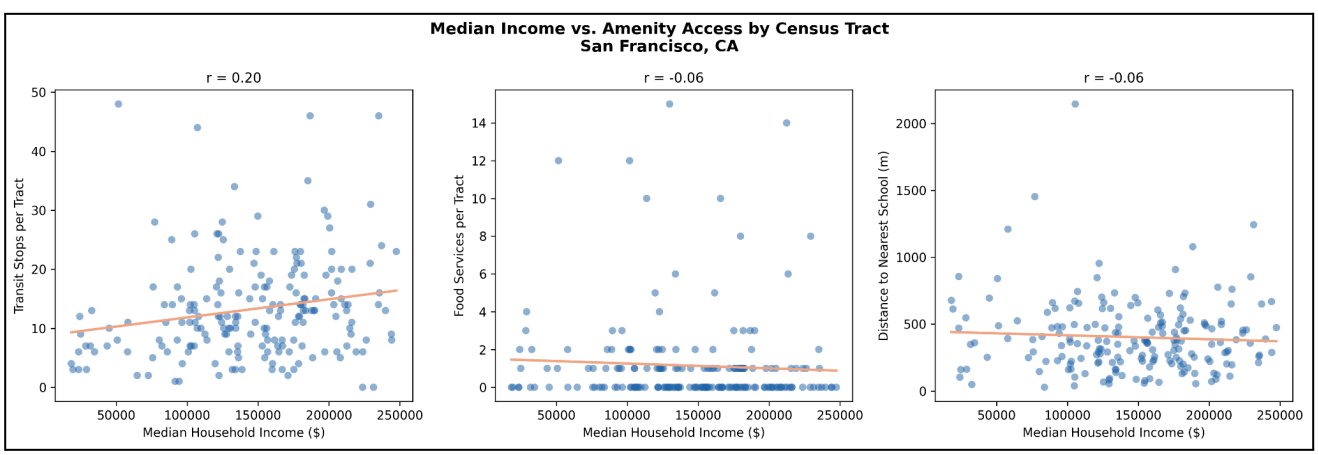

Figure 10. Correlation analysis of Median Income vs. Amenity Access by Census Tract.

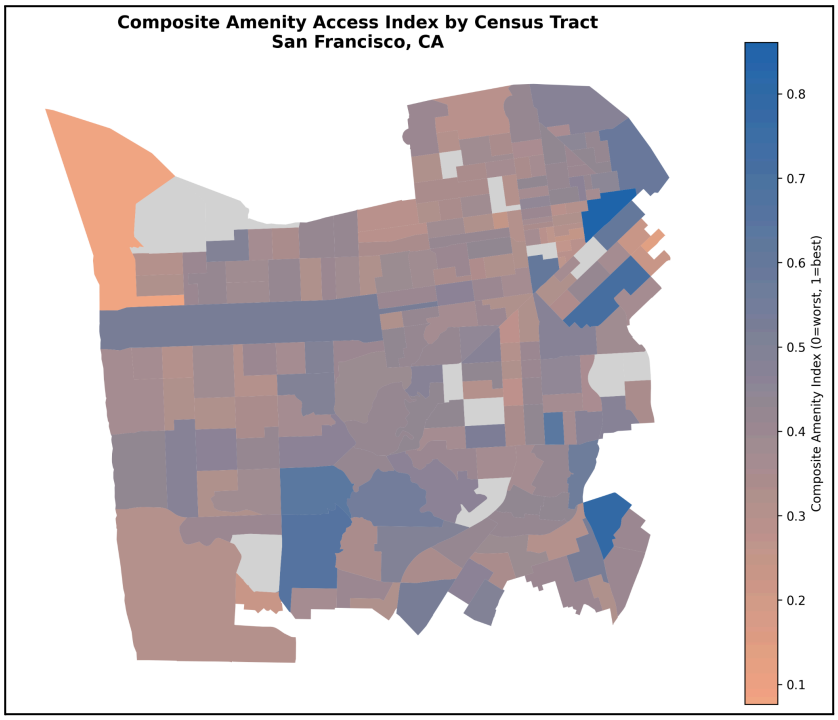

Figure 11. Map of Access Index by Census Tract.

6. Discussion and Limitations

Several limitations shape how these findings should be interpreted. Most fundamentally, the entire analysis depends on census tract boundaries, meaning an amenity just across a tract line does not get counted for the neighboring area even if residents can easily reach it, which may distort local patterns of access. The composite index also treats food, transit, and school access as equally important with no empirical basis for that choice, and different weighting decisions would produce a meaningfully different map. Statistically, the strongest correlation found was $r = 0.20$, explaining only about 4% of variance in transit access, which means income alone is a weak predictor and our findings should be read as exploratory rather than conclusive. Finally, 93% of median income values were suppressed by the Census Bureau for small or low-population tracts, meaning the communities most likely to experience disadvantage are also the least represented in the analysis.

7. Conclusion & Future Research

This project looked at whether lower-income neighborhoods in San Francisco face compounded disadvantages in access to food, transit, and schools. Overall, we found that income by itself does not strongly predict access to any single type of amenity. Some low-income neighborhoods still have nearby services, while some higher-income areas are not uniformly well served. However, when we combine food, transit, and school access into a single index, clearer patterns emerge. Certain parts of the city experience overlapping gaps across multiple services at once, especially in specific western and southeastern tracts. These patterns would not be visible if we only looked at one type of amenity at a time, which supports the idea that deprivation builds across different parts of everyday life rather than appearing in isolation.

For policy, this suggests that treating food access, transit access, and educational access as separate problems can miss where the biggest needs actually are. A more effective approach would focus on neighborhoods where multiple disadvantages overlap, rather than addressing each system independently. For example, improving transit connectivity, increasing high-quality food options, and strengthening school resources in the same areas could have a greater combined impact on opportunity.

More broadly, this project shows why it is important to measure urban inequality in a way that reflects real lived experience, where people rely on several systems at once. It also suggests future research could look more closely at how these overlapping disadvantages affect outcomes like health, mobility, and long-term opportunity over time.

References

Akbar, Prottoy A. 2024. "Public Transit Access and Income Segregation." CEPR Discussion Paper 19558.

Apparicio P, Cloutier MS, Shearmur R. The case of Montréal's missing food deserts: evaluation of accessibility to food supermarkets. Int J Health Geogr. 2007 Feb 12;6:4.

HAND, DAVID J. *Dark Data: Why What You Don't Know Matters*. Princeton University Press, 2020.

Krieger, Nancy. 2006. "A Century of Census Tracts: Health & the Body Politic (1906–2006)." Journal of Urban Health 83 (3): 355–61.

Maharjan, S., Janatabadi, F., & Ermagun, A. (2024). Spatial Inequity of Transit and Automobile Access Gap across America for Underserved Population. Transportation Research Record: Journal of the Transportation Research Board, 2678(1), 674-690.

Ramirez, Yesenia, "Food Insecurity in San Francisco (Examination of some local and state interventions)" (2022). *Master's Projects and Capstones*. 1485.

USDA. 2025. "Supplemental Nutrition Assistance Program (SNAP) | USDA-FNS." Usda.gov. February 20, 2025. https://www.fns.usda.gov/snap/supplemental-nutrition-assistance-program.

Zheng, Angela, and James Graham. 2022. "Public Education Inequality and Intergenerational Mobility." American Economic Journal: Macroeconomics 14 (3): 250–82.